

SES 251: Air and Atmospheric Sciences

Instructor: Kezhou (Melody) Lu, Ph.D.

Spring, 2024

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Class Hours: T/Th 9:25-10:40am

Office: ES&T 2108 at Georgia Tech

Phone: 510-820-7974

Office Hours: Virtual by appointment

Class Room: Tapley 130

Course Description

This course provides an introduction to the Earth's atmosphere and the Earth's climate system. The Earth's system consists of the atmosphere, the ocean, the land, and the cryosphere, which interact through physical, chemical, and biological processes. The course is designed to primarily focus on fundamental knowledge of the atmosphere and climate. The first half of the class will cover basic concepts about the atmosphere, including its structure and composition, radiative transfer, global energy balance, atmospheric thermodynamics, and large-scale atmospheric dynamics (e.g., Hadley Circulation), precipitation, and cloud formation. The second half will provide an overview of climate science, including climate variability, climate oscillations (e.g., El Niño), wind-driven ocean circulation, atmospheric teleconnection as well as atmosphere-ocean interaction.

Recommended Materials

- Course slides will be available on Canvas/files.
- Textbook 1: **Essentials of Meteorology, an invitation to the atmosphere**, 8th edition, by C. Donald Ahrens and Robert Henson.
- Textbook 2: **Introduction to Modern Climate Change**, 2nd edition, by Andrew Dessler
- Textbook 3: **Global Warming: Understanding the Forecast**, 2nd edition, by David Archer

Prerequisites/Corequisites

Prerequisites: Calculus I and Calculus II, Linear Algebra

How to Do Well?

Successful students:

1. Attend every class
2. Pay attention, take notes, and ask questions
3. Do well on quizzes and exams
4. Start lab assignments early
5. Study for quizzes and exams with a focus on understanding the concepts

Course Structure

Course Goals and Objectives

The goals of SES 251 Air and Atmospheric Sciences are for students to:

1. Understand the basic physical principles and meteorological terminology for weather and climate
2. Understand various aspects of atmosphere and climate, including solar radiation, global atmospheric circulation patterns, precipitation processes, climate systems and how to model the climate system

Labs/Quizzes/Exams

Lab Assignments

1. There will be six lab assignments throughout the semester. In general, lab assignments will be given on Thursday during lecture time. For each lab, you will have two weeks to complete. Please refer to the "Schedule and weekly learning goals" for deadlines of each lab assignments.
2. Collaboration is encouraged; however, **plagiarism is strictly prohibited. In the case of plagiarism, the lab will receive no credits.**
3. **The use of AI, including platforms like ChatGPT, to complete lab assignments is not allowed.** However, you are encouraged to utilize AI as a supplementary tool to enhance your comprehension of the class materials.

In-class Quizzes

1. Five quizzes will be conducted **in person** at the conclusion of each chapter. Quizzes primarily consist of multiple-choice questions and include a few short-answer questions.
2. **Quizzes are closed-book and must be completed independently in class.**
3. Only one make-up quiz will be given at end of the semester to cover for a missed quiz due to excused absence.
4. If you are unable to attend classes in person due to an excused absence, please inform me in advance. You will be provided with an online quiz as an alternative.

Group Project

1. The class will be broken into groups of two or three students. Each group will choose a topic related to atmosphere and climate. Some recommended topics will be announced later. You are encouraged to choose your own topic.
2. Each group will submit **one** written report at the end of the semester, **no later than 05/10**.
3. Each group will do an oral presentation for 15 minutes. Each group member will present a section of the research project.

Midterms and Final Exams

The two midterms and one final consists of multiple choices, short answer questions, graph interpretations, charts and essays. The exams will be a combination of factual recalls, conceptual understanding, and application questions.

Make-up Exams

The only acceptable reasons for rescheduling an exam are having a conflicting exam, medical emergency, jury summons, or military duty. No make-up exam will be offered for any other reason.

1. It is impossible to create an equivalent experience without placing the student at either an advantaged or disadvantaged status
2. No make-up exams are given unless you have a written excuse notifying me of special circumstances. These notifications usually come from the Dean's Office and are of a serious nature. You must inform me in advance, if you will be unable to take an exam at the scheduled date and time for the reasons stated above. All excuses must be documented!

3. No accommodation will be made after an exam is missed regardless of the reason if no communication occurs ahead of time!

Grading Policy

I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments, lab reports, exams and group project, using the following proportions:

- 45% of your grade will be determined by 3 in class exams (15% each).
- 20% of your grade will be determined by the group project that includes a written report and an oral presentation (10% each)
- 20% of your grade will be determined by lab reports
- 10% of your grade will be determined by quizzes
- 5% of your grade will be from your attendance

Grading Scale

A (95 – 100%), A⁻ (90 – 94%),
B⁺ (87 – 89%), B (83 – 86%), B⁻ (80 – 82%),
C⁺ (76 – 79%), C (70 – 75%), C⁻ (67 – 69%),
D⁺ (63 – 66%), D (60 – 62%),
F (0 – 59%)

Procedure for Disputing a Grade Policy

If you disagree with a grade on a test or assignment, you may submit a written request via email. Written requests must indicate the question(s) or assignment elements that are disputed AND include your argument for why you think you should have received a different grade. These requests must be made within 3-days of receiving the grade. Disagreement with the final course grade is handled through the College's official Grade Grievance process. Grievance must be filed in the Dean's Office by midterm of the following semester.

Course Policies

During Class

I understand that the electronic recording of notes will be important for class and so computers will be allowed in class. Please refrain from using computers for anything but activities related to the class. Phones are prohibited as they are rarely useful for anything in the course. Eating and drinking are allowed in class but please refrain from it affecting the course. Try not to eat your lunch in class as the classes are typically active.

Attendance Policy

Attendance is expected in all lecture and lab sections. Valid excuses for absence will be accepted before class. In extenuating circumstances, valid excuses with proof will be accepted after class. For every class missed the participation grade will be dropped 1 point.

Policies on Incomplete Grades and Late Assignments

If an extended deadline is not authorized by the instructor or department, an unfinished incomplete grade will automatically change to an F after either (a) the end of the next regular semester in which the student is enrolled (not including summer sessions), or (b) the end of 12 months if the student is not enrolled, whichever is shorter. Incompletes that change to F will count as an attempted course on transcripts. The burden of fulfilling an incomplete grade is the responsibility of the student.

Late assignments will be accepted for no penalty if a valid excuse is communicated to the instructor before the deadline. After the deadline, assignments will be accepted for a 50% deduction to the score up to 2 days after the deadline. After this any assignments handed in will be given 0.

Academic Integrity and Honesty

Students are expected to uphold the highest standards of academic integrity and honesty as described below. Any incident of academic dishonesty will immediately be reported to the Office of the Academic Dean. "At the heart of Spelman College's mission is academic excellence, along with the development of intellectual, ethical and leadership qualities. These goals can only flourish in an institutional environment where every member of the College affirms honesty, trust, and mutual respect. All members of the academic community of Spelman College are expected to understand and follow the basic standards of honesty and integrity, upholding a commitment to high ethical standards. Students are expected to read and abide by Spelman College Code of Conduct (see the current Spelman College Student Handbook) and are expected to behave as mature and responsible members of the Spelman College academic community. Students are expected to follow ethical standards in their personal conduct and in their behavior towards other members of the community. They are expected to observe basic honesty in their work, words, ideas, and actions. Failure to do so is a violation of the Spelman College Academic Integrity Policy. Violators will be subjected to the sanctions outlined in the Spelman College Bulletin. Some common violations of these basic standards of academic integrity include, but are not limited to: (1) Plagiarizing a written assignment. All statements of your work should be properly cited; (2) Copying and taking to one another during any exams; (3) Acting to facilitate copying during an exam; (4) Presenting others work as one's own; (5) Facilitating academic dishonesty. "

Accommodations for Disabilities and Pregnancy

Spelman College is sensitive to the special needs of students with disabilities. Any student who feels she may need an accommodation based on the impact of a disability should contact the Office of Disability Service at 404-270-5289 in Mac Vicar Hall to coordinate reasonable accommodations. The College recommends that any student who is pregnant and enrolled in any laboratory and/or physical activity get written permission from her attending physician before participating in any laboratory and/or physical activity for the course. Please refer to the Student Handbook (pg. 32) for the College's policy statement on Students with Serious Medical Conditions. To ensure that all laboratory safety measures are in place, students must get approval before embarking on their group projects. If the group projects will involve wet experiments, a Material Safety Data Sheet will be made available for any hazardous materials used in the project.

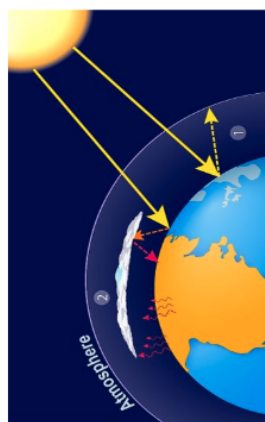
Student Access Statement

Spelman College is committed to ensuring the full participation of all students in its programs. If you have a documented disability (or think you may have a disability) and, as a result, need a reasonable accommodation to participate in class, complete course requirements, or benefit from the College's programs or services, you should contact Student Access Center (SAC) as soon as possible. The Student Access Center works with students confidentially and does not disclose any disability-related information without their permission. Contact the Student Access Center at 404-270-5289 for additional information.

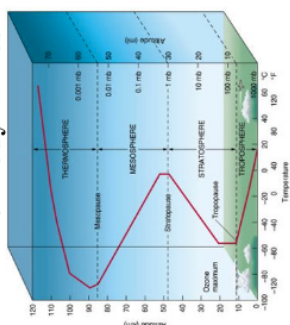
Course Outline

Fundamentals of the Earth's Atmosphere

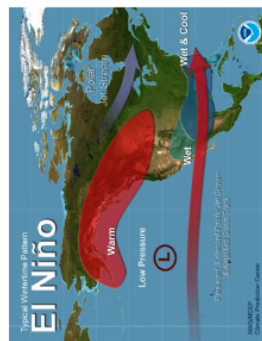
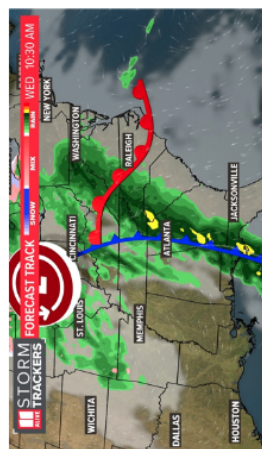
Radiation



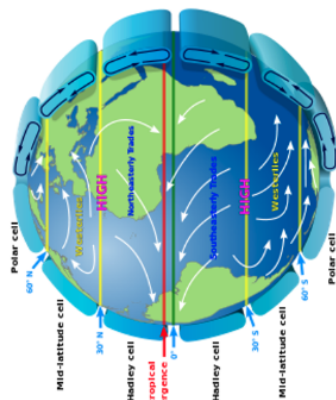
Thermodynamics



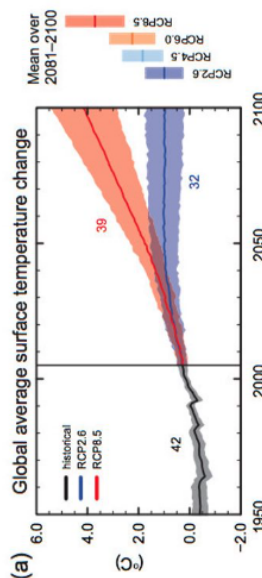
Weather vs Climate



Atmospheric Circulation (Dynamics)



Climate Change & Climate Modeling



Schedule and weekly learning goals

The schedule is tentative and subject to change. The learning goals below should be viewed as the key concepts you should grasp after each week, and also as a study guide before each exam, and at the end of the semester. Each exam will test on the material that was taught up until 1 week prior to the exam. The applications in the later half of the semester tend to build on the concepts in the first half of the semester though, so it is still important to at least review those concepts throughout the semester.

Week 01, 01/15 - 01/19: Chapter 0 - Introduction (Assigning Lab 1)

- The Atmosphere and the Scientific Method
- Weather, Climate and Meteorology
- Components of Earth's Atmosphere
- Vertical Structure of the Atmosphere

Week 02, 01/22 - 01/26: Chapter 1 - Atmospheric Radiation

- Temperature and Heat Transfer
- Radiant Energy
- Radiation-Absorption, Emission, and Equilibrium

Week 03, 01/29 - 02/02: Chapter 1 - Atmospheric Radiation (Assigning Lab 2 and Lab 1 due)

- Blackbody Radiation and Radiative Properties of the Atmosphere
- Layer Model: Solar Insolation, Energy Balance, and Greenhouse Effect
- Why Earth Has Seasons
- Radiative Forcing

Week 04, 02/05 - 02/09: Chapter 2 - Thermodynamics of the Atmosphere

- Temperature, Pressure and Ideal Gas Law
- Atmospheric Humidity
- Condensation: Dew, Fog, and Clouds

Week 05, 02/12 - 02/16: Chapter 2 - Thermodynamics of the Atmosphere (Assigning Lab 3 and Lab 2 due)

- Atmospheric Stability
- Calculating Atmospheric Stability

Week 06, 02/19 - 02/23: Chapter 2 - Thermodynamics of the Atmosphere

- Stability and Cloud Development
- Precipitation

Week 07, 02/26 - 03/01: Chapter 3 - Dynamics of the Atmosphere (Assigning Lab 4 and Lab 3 due)

- Atmospheric Pressure
- Surface and Upper-level Winds

Week 08, 03/04 - 03/08: Chapter 3 - Dynamics of the Atmosphere and **Exam 1 on 03/07**

- Why the Wind Blows

Week 09, 03/11 - 03/15: **Spring Break**

Week 10, 03/18 - 03/22: Chapter 3 - Dynamics of the Atmosphere (**Assigning Lab 5 and Lab 4 due**)

- Scales of Atmospheric Motion
- Eddies
- Local Wind Systems

Week 11, 03/25 - 03/29: Chapter 3 - Dynamics of the Atmosphere

- Global Winds
- Atmosphere-Ocean Interactions

Week 12, 04/01 - 04/05: Chapter 3 - Dynamics of the Atmosphere (**Assigning Lab 6 and Lab 5 due**)

- Atmosphere-Ocean Interactions
- Air Masses and Fronts
- Mid-latitude Cyclones

Week 13, 04/08 - 04/12: Chapter 4 - Climate Change and Modeling **Exam 2 on 04/09**

- The Concept of Feedback: Water Vapor Feedback, Ice Albedo Feedback, Cloud Feedback, Lapse Rate Feedback

Week 14, 04/15 - 04/19: Chapter 4 - Climate Change and Modeling (**Lab 6 due**)

- Carbon Cycle: Source and Sink, Carbon Reservoirs, Atmosphere-ocean-land Cycle
- Global Warming
- Evidence of Global Warming: Temperature measurements, Paleoclimatology, Solar Forcing, Internal Variability

Week 15, 04/22 - 04/26: Chapter 4 - Climate Change and Modeling

- Climate Modeling
- Impacts of Climate Change: Changes in Hydrological Cycle, Snowpack and Extreme Weather

Week 16, 04/29 - 05/03: Project Presentation and Exam Review

- [Group Presentation](#) (Tentative Schedule)
- Exam Review (Tentative Schedule)

Week 17, 05/06 - 05/10: **Final Exam on 05/09**